UNIVERSITY OF TORONTO

Precise Pop Cans

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MAT186

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1.0 Introduction

The optimization of a soda can is a common problem given to first-year students. These types of questions usually consist of finding the ideal dimensions of a cylinder such that material use is minimized. We can find the formula to find the optimal soda can using the following:

This is the expression of the volume of a cylinder, which can be rearranged to $h = \frac{V}{\pi r^2}$. The surface area including the top and bottom is given by $s = 2\pi rh + 2\pi r^2$. Since, $h = \frac{V}{\pi r^2}$ the surface area, *S*, can be re-expressed as $s = 2Vr^{-1} + 2\pi r^2$. To find the value of *r* that will give the smallest surface area we find the derivative, set it equal to zero and solve for *r*

$\frac{dS}{dr} = -2Vr^{-2} + 4\pi r$. This will equal zero when $r = \sqrt[3]{\frac{V}{2\pi}}$ and substituting into the expression above $h = \sqrt[3]{\frac{4V}{\pi}}$ then $\frac{h}{r} = \sqrt[3]{\frac{4V}{\frac{\pi}{\frac{V}{2\pi}}}} = 2$, so $h = 2r$. [5]

This tells us that the optimal dimensions for a soda can are when the height is equal to its diameter (2R). Studies have shown that when manufactured cans are measured, their dimensions slightly vary. This can be due to a variety of reasons, such as; Airspace, material thickness, Joinery, Cutting, and Human Factors. This report will examine why this is the case, and incorporate math into the reasoning.

2.0 Airspace

Due to the fact that air-tight soda cans are not entirely filled with liquid, the excess air in a soda can lead to pressure build-up when comparing the volume of air to the volume of liquid. We can calculate the amount of pressure in the liquid versus air using $P = \frac{F}{A}$. To be more specific, we can

use the ideal gas law arranging for pressure, $P = \frac{nRT}{v}$ and the fluid pressure formula $P = \rho gh$. By finding the pressure on the can, we can determine whether it is large enough to affect the dimensions of the can [4].

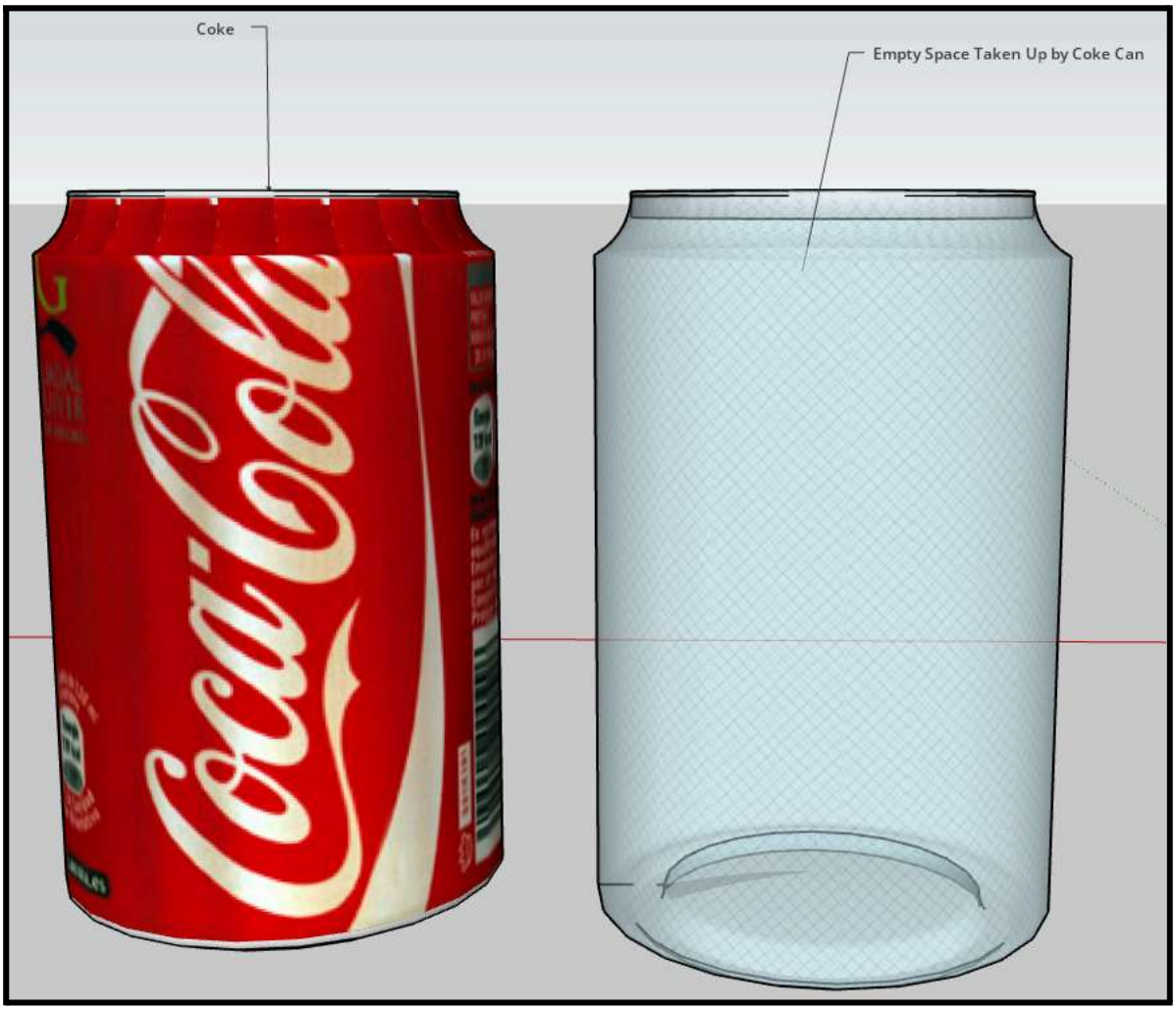

Figure 1. Visual comparison of the total area to the airspace of a Coke can (differs based on the thickness of walls).

3.0 Material Thickness

Dimensions can be affected by material thickness because the density, material thickness, and weight of the soda can affect the surface area, which is related to the dimensions of the soda can. The

figure below shows the formula for testing the dimensions of the soda can [3]. The top and bottom of the cans tend to be thicker than the walls to withstand the pressure from inside the can. However, the sides are thin to allow for it to be easily carried and to reduce the material costs.

$$\text{density} \approx \frac{\text{surface area} \times \text{thickness}}{\text{weight}}$$

4.0 Joinery

Soda cans are sealed using a crimping process where the lid of the can is placed and sealed using a soda can sealer. This crimping process may add or reduce the height of the can making the sizes of cans vary. Since this can affect the height of a can, a study can be conducted using the derived optimization equation h=2r to see how the cans differ due to joinery from the error [2].

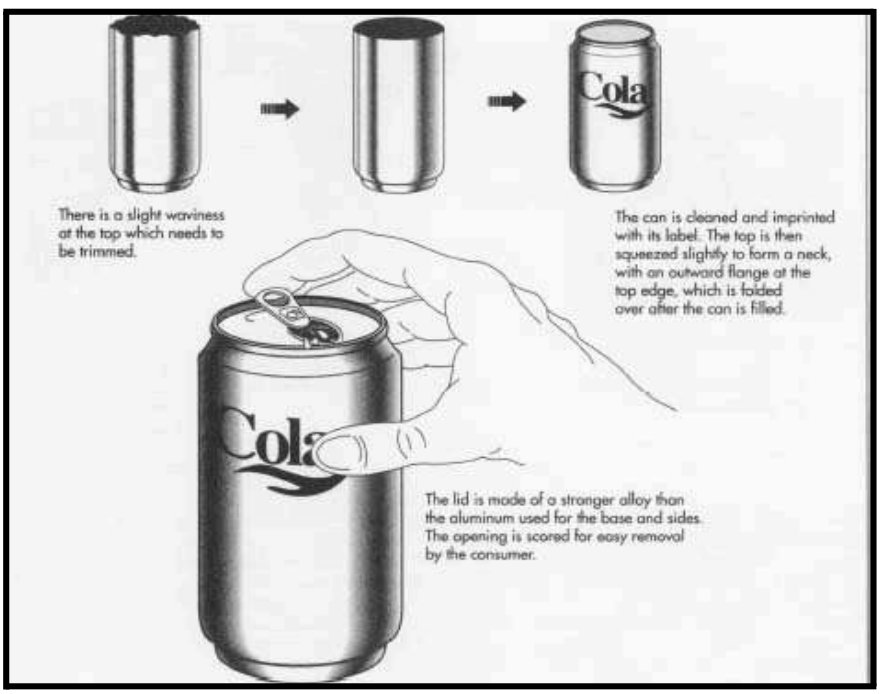

Figure 2. Visual description of the connection and labelling of a Coke can [1]

5.0 Cutting

The cutting of aluminum when manufacturing soda cans must be precise, or the diameter-to-height ratio will be off. It is important to consider the different ways aluminum sheets

are cut, material wastage, and precision and how these impact the problem. While companies try to minimize material wastage, the dimensions of the can may be affected since they want to make the most of the available material. If the radius of the top increases, the length of the sides (cut into rectangles) would have to increase too. On the other hand, if the radius decreases, the height of the sides would have to increase to accommodate the required volume of liquid and pressurized air.

Figure 3. Wide/increased radius cans vs. narrow/decreased radius cans

6.0 Human Factors

Coke cans are not normally created to be mathematically optimal. The aspect of the real world prioritizes alternate factors above the need to optimize space. Because of things like ergonomics, marketing, and practicality, real-world Coke cans are not mathematically optimal.

6.1 Ergonomics and Consumer Preference

The cans are made to be easily handled and gripped. Shorter cans are more stable, while taller cans are simpler to hold. Dimensions that minimize surface area and satisfy $h = 2r$ are

mathematically optimal. However, because of handling issues, actual cans are different from this.

6.2 Branding and Visual Appear

Smaller cans work well with traditional soda cans, whereas taller cans are utilized for high-end goods. Their shape also affects how readable labels and logos are. The sizes can also differ based on the country. In some countries, they may prefer shorter and wider cans, while others prefer taller and narrower cans. A cylindrical can's surface area can be calculated using:

$$A = 2\pi rh + 2\pi r^2$$

This affects the quantity of room available for marketing and branding design.

6.3 Shelf and Storage Efficiency

Wider cans maximize stability and stackability during storage, whereas thinner cans fit better in vending machines and conserve shelf space. The restriction on volume:

$$V = \pi r^2 h$$

is modified to replace purely mathematical optimization with storage and transportation restrictions.

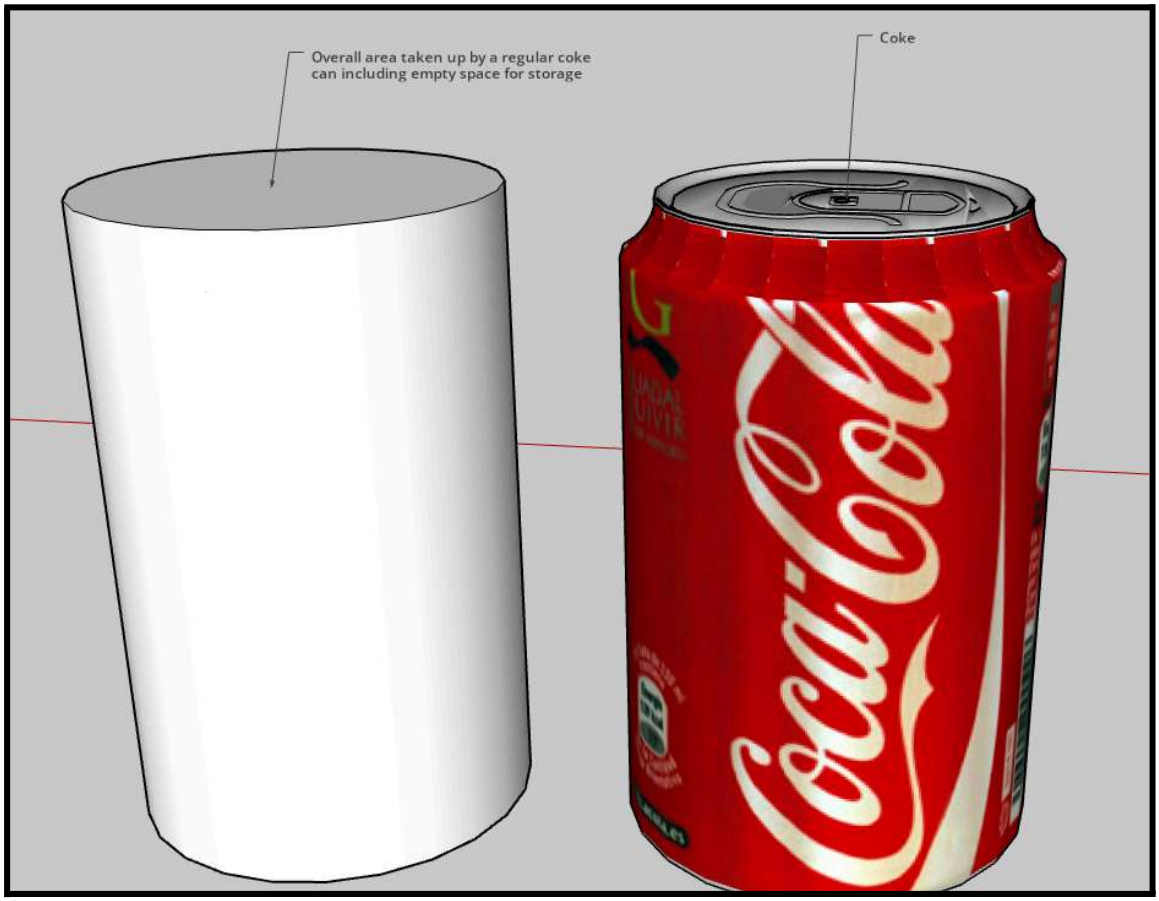

Figure 4. Representation of the overall packing space of a Coke can

7.0 Conclusion

To conclude this report, it was found that air space, material thickness, joinery and cutting, all impact the dimensions of a soda can. We found the reasoning for these and related them to possible mathematical tests. Airspace can be related to pressure, thickness can be related to density, and Joinery can be related to the optimization equation for soda cans h=2R. When analyzing the human impacts, the size of soda cans causes trade-offs from mathematical optimization and involves trade-offs between customer comfort, marketing, efficiency, and sustainability. Although mathematical theory serves as a basis, final designs are determined by practical factors.

8.0 References

[1] madehow, "Aluminum beverage can," How aluminum beverage can is made - material, production process, manufacture, making, used, composition, structure, https://www.madehow.com/Volume-2/Aluminum-Beverage-Can.html (accessed Mar. 9, 2025).

[2] L. P. Mmldigi, "How are cans sealed? / how to seal cans? - levapack," www.levapack.com, https://www.levapack.com/how-aluminum-cans-are-sealed/ (accessed Mar. 9, 2025).

[3] Illustrative Mathematics, "How thick is a soda can? variation II," Illustrative Mathematics, https://tasks.illustrativemathematics.org/content-standards/tasks/1192 (accessed Mar. 9, 2025).

[4] University of Colorado, "G430: Pressure and temperature – the collapsing can," Lecture Demonstration Manual General Chemistry, https://www.colorado.edu/lab/lecture-demo-manual/g430-pressure-and-temperature-collapsing-can#:~:text=The%20cold%20water%20will%20also,causes%20the%20can%20to%20collapse (accessed Mar. 9, 2025).

[5] L. McMullin et al., "Soda Cans," Teaching Calculus, https://teachingcalculus.com/2015/05/13/soda-cans/ (accessed Mar. 9, 2025).

[6] 3D Design Software, 3D Modeling & Drawing, Sketchup, www.sketchup.com/en. (accessed 9 Mar. 2025.)